

Christos Koutras

Research Interests

Data Integration, Schema and Data Matching, Metadata Generation

Education

- 2018–2024 **PhD, Computer Science**, *Delft University of Technology (TU Delft)*, Netherlands, Expected Graduation: Fall 2024.
Research Topic: Tabular Schema Matching for Modern Settings
Supervisors: Asterios Katsifodimos, Christoph Lofi, Geert-Jan Houben
- 2016–2018 **Master of Philosophy (MPhil), Computer Science**, *The Hong Kong University of Science & Technology (HKUST)*, Hong Kong.
Research Topic: Databases, Data Management
Thesis: Query Estimation Structures in Geo-Social Networks
Supervisor: Dimitris Papadias
GPA: 3.94/4
- 2009–2015 **Diploma (M.Eng), Electrical and Computer Engineering**, *National Technical University of Athens (NTUA)*, Greece.
GPA : 8.24/10
Major: Computer Science
Thesis: Monitoring Continuous k -Nearest Neighbor Queries Over Moving Uncertain Objects
Supervisors : Kostas Patroumpas, Yannis Vasileiou

Experience

- November 2024 - Present **Postdoctoral Research Associate**, *New York University*, Brooklyn, NY, USA.
Visualization and Data Analytics (VIDA) Research Center, Tandon School of Engineering
- January - April 2026 **Course Section Leader**, *New York University*, Manhattan, NY, USA.
DS-GA 3001 Data Engineering
Responsible for organizing and delivering the course's lab sessions.
- December 2023 - March 2024 **Applied Scientist Intern**, *Amazon Web Services*, SF Bay Area, USA.
Worked on optimizing LLM-usage in Data Wrangling pipelines
- April–July 2022 **Applied Scientist Intern**, *Amazon Web Services*, SF Bay Area, USA.
Internship with the AWS GraphML team, where I worked towards self-supervised join discovery among tabular data.
- November–December 2018–2023 **Course Coordinator**, *Delft University of Technology*, Netherlands.
CSE1500: Web and Database Technology
Responsible for organizing the course's DB lab sessions and teaching tutorials, while also for designing the DB assignments.
- February–April 2018–2023 **Course Coordinator**, *Delft University of Technology*, Netherlands.
CSE1505: Information and Data Management
Responsible for organizing the course's lab sessions and teaching tutorials. Also, for designing and grading lab assignments and exam tasks.

- February– **Teaching Assistant**, *The Hong Kong University of Science & Technology*, Hong Kong.
 May COMP 3311: Database Management Systems
 2018 Responsible for teaching the tutorials of the second half of the course on topics such as : Functional Dependencies and Normalization, Memory Hierarchy and File Structures, Indexing and Hashing, Join Algorithms, Query Processing and Optimization
- September– **Teaching Assistant**, *The Hong Kong University of Science & Technology*, Hong Kong.
 December COMP 3711: Design and Analysis of Algorithms
 2017 Responsible for designing and grading the course assignments on Greedy Algorithms, Dynamic Programming, Minimum Spanning Trees, Shortest Path Algorithms
- July–August **Visiting Student**, *Oxford University*, United Kingdom.
 2017 Worked on Query Optimization with Access Patterns under Integrity Constraints in collaboration with George Konstantinidis
- February– **Teaching Assistant**, *The Hong Kong University of Science & Technology*, Hong Kong.
 May COMP 3311: Database Management Systems
 2017 Responsible for teaching the tutorials of the first half of the course on topics such as: ER diagrams, Relational Data Model, Relational Algebra, SQL, Functional Dependencies and Normalization

Professional Service

External Reviewer.

VLDB 2023/2024, IEEE ICDE 2019/2023, EDBT 2024, ACM SIGMOD 2018/2021, ACM SIGSPATIAL 2017, SSTD 2017

PC Member.

DBML 2022-2024/2026 (Workshop in conjunction with ICDE), ICDE 2024 Demonstration Track, ECML/PKDD 2024 Applied Data Science Track, VLDB 2026/2027, ICDE 2026/2027, SIGMOD 2026

Reviewer for Journals.

Data & Knowledge Engineering (DKE), Patterns (Cell Press), Information Systems

Publications

- SIGMOD 2026 **BDIViz in Action: Interactive Curation and Benchmarking for Schema Matching Methods**, *To appear in SIGMOD 2026*, Bangalore, India.
 Eden Wu, **Christos Koutras**, Juliana Freire
- SIGMOD 2026 **BDI-Kit Demo: A Toolkit for Programmable and Conversational Data Harmonization**, *To appear in SIGMOD 2026*, Bangalore, India.
 Roque Lopez, Yurong Liu, **Christos Koutras**, Juliana Freire
- Patterns 2026 **BDI-Kit: An AI-powered toolkit for biomedical data harmonization**, *Patterns*.
 Roque Lopez, Aecio Santos, **Christos Koutras**, Juliana Freire
- VLDB 2025 **OmniMatch: Joinability Discovery in Data Products**, *In Proceedings of VLDB 2025*, London, UK.
Christos Koutras, Jiani Zhang, Xiao Qin, Chuan Lei, Vasileios Ioannidis, Christos Faloutsos, George Karypis, Asterios Katsifodimos
- IEEE VIS 2025 **BDIViz: An Interactive Visualization System for Biomedical Schema Matching with LLM-Powered Validation**, *In Proceedings of IEEE International Conference on Visualization 2025*, Vienna, Austria.
 Eden Wu, Dishita G Turakhia, Guande Wu, **Christos Koutras**, Sarah Keegan, Wenke Liu, Beata Szeitz, David Fenyo, Claudio Silva, Juliana Freire
- IEEE Data Engineering Bulletin **Large Language Models for Data Discovery and Integration: Challenges and Opportunities**, *In IEEE Data Engineering Bulletin*.
 Juliana Freire, Grace Fan, Benjamin Feuer, **Christos Koutras**, Yurong Liu, Eduardo Pena, Aecio Santos, Claudio Silva, Eden Wu
- TRL@ NeurIPS 2022 **SiMa: Effective and Efficient Matching Across Data Silos Using Graph Neural Networks**, *First Table Representation Learning Workshop @ NeurIPS 2022*, New Orleans, Louisiana, USA.
Christos Koutras, Rihan Hai, Kyriakos Psarakis, Marios Fragkoulis, Asterios Katsifodimos

- TKDE **Data Lakes: A Survey of Functions and Systems**, *Transactions on Knowledge and Data Engineering*.
Rihan Hai, **Christos Koutras**, Christoph Quix, Matthias Jarke
- ICDE 2023 **Amalur: Data Integration Meets Machine Learning**, *In Proceedings of IEEE ICDE 2023*, Anaheim, California, USA.
Rihan Hai, **Christos Koutras**, Andra Ionescu, Ziyu Li, Wenbo Sun, Jessie van Schijndel, Yan Kang, Asterios Katsifodimos
- CIDR 2022 **Amalur: Next-generation Data Integration in Data Lakes**, *Conference on Innovative Data Systems Research (CIDR) 2022*.
Rihan Hai, **Christos Koutras**, Andra Ionescu, Marios Fragkoulis, Asterios Katsifodimos
- VLDB 2021 **Valentine in Action: Matching Tabular Data at Scale**, *In Proceedings of VLDB 2021*, Copenhagen, Denmark.
Christos Koutras, Kyriakos Psarakis, George Siachamis, Andra Ionescu, Marios Fragkoulis, Angela Bonifati, Asterios Katsifodimos
- IEEE ICDE 2021 **Valentine: Evaluating Matching Techniques for Dataset Discovery**, *In Proceedings of IEEE ICDE 2021*, Chania, Greece.
Christos Koutras, George Siachamis, Andra Ionescu, Kyriakos Psarakis, Jerry Brons, Marios Fragkoulis, Christoph Lofi, Angela Bonifati, Asterios Katsifodimos
- SeaData Workshop 2020 **REMA: Graph Embeddings-based Relational Schema Matching**, *In Workshop Proceedings of the EDBT/ICDT 2020 Joint Conference*, Copenhagen, Denmark.
Christos Koutras, Marios Fragkoulis, Asterios Katsifodimos, Christoph Lofi
- VLDB Workshop 2019 **Data as a Language: A Novel Approach to Data Integration**, *Proceedings of the VLDB 2019 PhD Workshop*, Los Angeles, California, USA.
Christos Koutras
- SSDBM 2017 **Probabilistic k-Nearest Neighbor Monitoring of Moving Gaussians**, *Proceedings of the 29th International Conference on Scientific and Statistical Database Management*, Chicago, Illinois, USA.
Kostas Patroumpas, **Christos Koutras**

Projects & Technical Skills

- Valentine package Core maintainer of `valentine` python package. With Valentine users can find matches among columns of pandas DataFrames.
- Programming languages Ability to develop and implement algorithms in JAVA, C/C++ and Python. Familiarity with MATLAB, PostGre SQL and Git/GitHub.

Awards

- 2026 **IEEE ICDE Distinguished PC Award**.
Awarded to PC Committee members for outstanding contributions in reviewing and discussions.

Certifications

- 2023 **University Teaching Qualification**.
The University Teaching Qualification (UTQ) certifies the didactic competences of lecturers in academic education across the Netherlands. Recognized by all fourteen Dutch Universities.

Languages

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| Greek | Native Speaker | |
| English | Professional Working Proficiency | <i>CPE Certification C2, University of Michigan</i> |
| German | Elementary Proficiency | <i>Zertifikat B1, Goethe Institut</i> |